

Preparation of Papers for IEEE Sponsored Conferences & Symposia*

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I. INTRODUCTION

Wide-angle central cameras, such as fisheye cameras, capture a larger angular extent of the scene within a single image, providing broader visual coverage for geometric vision tasks such as visual odometry and 3D reconstruction. However, fully exploiting these additional observations requires an accurate geometric calibration over the entire image domain, especially in peripheral regions where the projection behavior changes significantly. Compared with conventional cameras, wide-angle cameras introduce two major challenges in calibration: First, the strong nonlinearity of the projection model makes model selection, parameter initialization, and nonlinear optimization more difficult [?], [?]. Second, severe projection variation and image deformation near the image boundary reduce the reliability of target detection and feature localization [?]. Accordingly, existing work has largely proceeded along two complementary directions: designing more suitable wide-angle camera models [?], [1], and improving the quality of calibration observations through target design, feature refinement, and view or observation selection [?], [?], [?].

This work focuses on the latter direction and investigates how planar calibration targets can provide more effective geometric constraints. Camera calibration typically estimates camera intrinsics and target poses from image observations of a target with known geometry. Planar targets such as checkerboards and AprilGrids are widely used due to their simple structures and convenient feature detection [?], [?]. For conventional lenses, translating and rotating a single planar target can provide sufficient observations to constrain the camera model. However, wide-angle cameras are more sensitive to the distribution of calibration observations. When observations are limited to a narrow radial range, the focal scale and projection-shape parameters can induce similar residual variations, leading to poorly separated

parameter directions and ill-conditioned linearized updates [?]. Therefore, wide-angle calibration requires not only a sufficient number of features, but also observations spanning different radial regions to provide more discriminative geometric constraints for camera model estimation.

However, realizing broad radial coverage with a single planar target is difficult in practice. In each view, a single target occupies only a limited and continuous image region, and its radial coverage is bounded by its projected extent. Enlarging the target projection increases the radial span, but also forces the same pattern to cross image regions with substantially different projection scales and distortion levels, placing greater demands on feature detection and subpixel localization. Existing wide-angle calibration methods mitigate this problem by using an intermediate camera model to reproject the target and adaptively refine missing or distorted features. Nevertheless, a practical trade-off remains between radial coverage and local observation reliability. This motivates a calibration framework that distributes reliable observations across different field-of-view regions while preserving stable local feature detection.

Motivated by these findings, we construct a distributed multi-board target using multiple boards, each carrying a uniquely coded AprilTag. The boards can be independently detected and associated within the same image, allowing observations to be distributed across different spatial and radial regions without requiring a single large continuous target to span the strongly distorted image domain. Each tag directly provides four sparse observations, referred to as outer corners. Although these distributed outer corners are sufficient to bootstrap an intermediate camera model, they remain locally sparse for final calibration. We therefore use the intermediate model to predict the image locations of predefined internal lattice points and refine them using local image evidence. In this way, spatially distributed but locally sparse outer-corner observations are extended into dense and geometrically consistent observations across different field-of-view regions. Based on this design, we develop a multi-board calibration system for wide-angle central cameras that integrates distributed target detection, dense observation recovery, and camera calibration into a unified pipeline.

Our main contributions are summarized as follows:

- We present an open-source, easy-to-deploy, and flexible multi-board calibration system for wide-angle

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central cameras.

- To address insufficient spatial coverage of calibration observations and unreliable feature refinement, we propose a distributed multi-board design based on single AprilTags, together with an intermediate-model-assisted internal lattice-point refinement method. On the evaluated sequence, the distributed design improves occupied-grid coverage by 14.6 percentage points under the same frame-board budget.
- We conduct a controlled analysis that isolates the contribution of recovered internal lattice observations under an identical Outer4 initialization. The results show that local observation densification provides complementary constraints for more faithful estimation of the wide-angle projection model.

Tests on several datasets and camera models cover reprojection error, ablations, cross-backend calibration, and held-out pose estimation. The results show reliable calibration and competitive accuracy.

II.

A. Selecting a Template (Heading 2)

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III. METHOD

A. Multi-Board Target Formulation

Our calibration target consists of N rigid boards, each carrying a uniquely coded AprilTag:

$$\mathcal{B} = \{B_0, B_1, \dots, B_{N-1}\}, \quad (1)$$

where B_0 is selected as the reference board. Let $\mathcal{F}_{B_b}$ denote the local coordinate frame of board B_b , and let $\mathbf{p}_{b,j} \in \mathbb{R}^3$ denote the j -th target point expressed in $\mathcal{F}_{B_b}$. The target points include the directly detected outer corners and the internal lattice points recovered in Sec. III-C.

We use $\mathcal{F}_{B_0}$ as the common coordinate frame of the multi-board target and fix $\mathbf{T}_{B_0 B_0} = \mathbf{I}$. Let $\mathbf{T}_{B_0 B_b} \in SE(3)$ denote the static rigid transform from $\mathcal{F}_{B_b}$ to $\mathcal{F}_{B_0}$. For frame t , let $\mathcal{F}_{C_t}$ be the camera coordinate frame and let $\mathbf{T}_{C_t B_0} \in SE(3)$ denote the pose of the reference board

in $\mathcal{F}_{C_t}$. A point on board B_b is therefore expressed in the camera frame as

$$\mathbf{x}_{t,b,j} = \mathbf{T}_{C_t B_0} \mathbf{T}_{B_0 B_b} \mathbf{p}_{b,j}. \quad (2)$$

Given the camera parameters $\boldsymbol{\theta}$ and the projection function $\pi(\cdot; \boldsymbol{\theta})$, the predicted image location is

$$\hat{\mathbf{u}}_{t,b,j} = \pi(\mathbf{x}_{t,b,j}; \boldsymbol{\theta}) = \pi(\mathbf{T}_{C_t B_0} \mathbf{T}_{B_0 B_b} \mathbf{p}_{b,j}; \boldsymbol{\theta}). \quad (3)$$

The corresponding image observation is denoted by $\mathbf{u}_{t,b,j} \in \mathbb{R}^2$. We collect all valid observations in

$$\mathcal{O} = \{(t, b, j, \mathbf{u}_{t,b,j})\}. \quad (4)$$

The unknown variables comprise the camera parameters $\boldsymbol{\theta}$, the per-frame reference-board poses $\{\mathbf{T}_{C_t B_0}\}$, and the static inter-board transforms $\{\mathbf{T}_{B_0 B_b}\}_{b=1}^{N-1}$.

B. Camera Model Initialization

At the beginning of calibration, no reliable camera model is available, while a usable initialization is important for nonlinear wide-angle calibration. The distributed AprilTag boards provide robust outer-corner observations across different image and radial regions. We use these observations to initialize the camera model. For a given camera-model family, we construct an initial parameter vector $\boldsymbol{\theta}^0$ using the image resolution and model-specific default values.

For board B_b observed in frame t , let $\mathcal{C}_{t,b}^{\text{out}}$ denote its valid outer-corner observations. We define the reprojection residual as

$$\mathbf{r}_{t,b,j}^{\text{out}}(\boldsymbol{\theta}, \mathbf{T}) = \mathbf{u}_{t,b,j} - \pi(\mathbf{T} \mathbf{p}_{b,j}; \boldsymbol{\theta}). \quad (5)$$

Given $\boldsymbol{\theta}^0$, a temporary frame-board pose is initialized by

$$\tilde{\mathbf{T}}_{C_t B_b}^0 = \arg \min_{\mathbf{T} \in SE(3)} \sum_{j \in \mathcal{C}_{t,b}^{\text{out}}} \|\mathbf{r}_{t,b,j}^{\text{out}}(\boldsymbol{\theta}^0, \mathbf{T})\|_2^2. \quad (6)$$

We perform this initialization independently for all valid frame-board observations and reject estimates with invalid poses or excessive reprojection errors. Let $\mathcal{S}$ denote the retained frame-board pairs. We then jointly refine the camera parameters and temporary poses:

$$\begin{aligned} & \boldsymbol{\theta}^*, \left\{ \tilde{\mathbf{T}}_{C_t B_b}^* \right\} \\ &= \arg \min_{\boldsymbol{\theta}, \{\tilde{\mathbf{T}}_{C_t B_b}\}} \sum_{(t,b) \in \mathcal{S}} \sum_{j \in \mathcal{C}_{t,b}^{\text{out}}} \rho \left(\left\| \mathbf{r}_{t,b,j}^{\text{out}}(\boldsymbol{\theta}, \tilde{\mathbf{T}}_{C_t B_b}) \right\|_2^2 \right). \end{aligned} \quad (7)$$

where $\rho(\cdot)$ is a robust loss function. At this stage, the boards are not coupled through the unknown inter-board geometry; each valid frame-board observation is assigned an independent pose. The optimized parameters $\boldsymbol{\theta}^*$ initialize the subsequent complete multi-board calibration and serve as the intermediate camera model for internal-point recovery.

C. Internal Lattice Point Recovery

The outer corners provide stable board localization and are sufficient to bootstrap an intermediate camera model. However, the four observations from each tag remain locally sparse for final calibration. We therefore use the intermediate model to recover image observations for predefined internal lattice points, combining model-guided prediction with local image evidence.

Given the intermediate camera parameters θ^* , we reuse the temporary frame-board pose $\tilde{\mathbf{T}}_{C_t B_b}^*$ obtained in Sec. ???. For an internal lattice point $\mathbf{p}_{b,j}$, $j \in \mathcal{C}_b^{\text{int}}$, its direct geometric projection

$$\hat{\mathbf{u}}_{t,b,j}^{\text{nom}} = \pi\left(\tilde{\mathbf{T}}_{C_t B_b}^* \mathbf{p}_{b,j}; \theta^*\right) \quad (8)$$

provides a nominal prediction and is used as a fallback when the boundary-based seed described below is unavailable.

To construct the primary seed, we sample support points along the four observed tag boundaries and back-project them onto the unit sphere using the intermediate camera model. The resulting top, bottom, left, and right boundary-ray curves are denoted by $\mathbf{r}_T(s)$, $\mathbf{r}_B(s)$, $\mathbf{r}_L(t)$, and $\mathbf{r}_R(t)$, respectively. Let $(s_j, t_j) \in [0, 1]^2$ be the normalized horizontal and vertical coordinates of $\mathbf{p}_{b,j}$ on the board lattice. We interpolate the opposite boundary curves as

$$\begin{aligned} \mathbf{r}_v &= \text{slerp}(\mathbf{r}_T(s_j), \mathbf{r}_B(s_j), t_j), \\ \mathbf{r}_h &= \text{slerp}(\mathbf{r}_L(t_j), \mathbf{r}_R(t_j), s_j). \end{aligned} \quad (9)$$

The two interpolated rays are blended and projected back to the image:

$$\tilde{\mathbf{d}}_{t,b,j}^{\text{int}} = \frac{\mathbf{r}_v + \mathbf{r}_h}{\|\mathbf{r}_v + \mathbf{r}_h\|_2}, \quad \tilde{\mathbf{u}}_{t,b,j}^{\text{int}} = \pi\left(\tilde{\mathbf{d}}_{t,b,j}^{\text{int}}; \theta^*\right). \quad (10)$$

Starting from $\tilde{\mathbf{u}}_{t,b,j}^{\text{int}}$, we perform a local search in the viewing-ray domain and then refine the candidate to subpixel accuracy in the original image. A recovered observation is rejected if it falls outside the image, lacks sufficient local image evidence, or deviates from its seed by more than a predefined threshold. Each accepted observation retains its board and lattice-point indices and is added to $\mathcal{O}$ for joint optimization together with the outer corners.

D. Spherical Bundle Adjustment

Standard bundle adjustment minimizes reprojection errors in the image plane. For wide-angle cameras, however, the ray deviation corresponding to a pixel residual depends on the image location and the local Jacobian of the projection model. We therefore provide optional ray-domain and hybrid objectives for final refinement.

Given an image observation $\mathbf{u}_{t,b,j}$, we back-project it using the current camera parameters and normalize it to obtain the observed ray:

$$\mathbf{d}_{t,b,j}^{\text{obs}} = \frac{\pi^{-1}(\mathbf{u}_{t,b,j}; \theta)}{\|\pi^{-1}(\mathbf{u}_{t,b,j}; \theta)\|_2}. \quad (11)$$

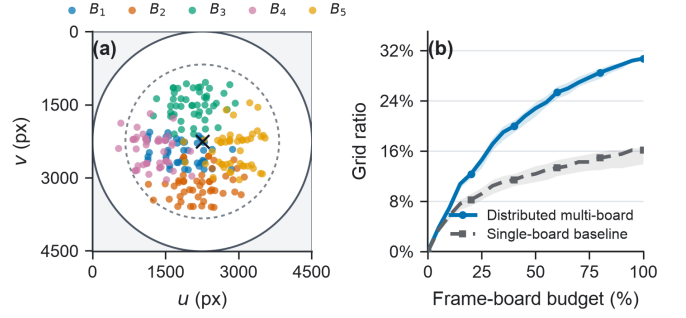


Fig. 1. xxxxxx

The corresponding predicted ray is obtained from the transformed target point defined in Sec. IV-A:

$$\mathbf{d}_{t,b,j}^{\text{pred}} = \frac{\mathbf{x}_{t,b,j}}{\|\mathbf{x}_{t,b,j}\|_2}. \quad (12)$$

Let $\mathbf{B}_{t,b,j} \in \mathbb{R}^{3 \times 2}$ be an orthonormal tangent basis at $\mathbf{d}_{t,b,j}^{\text{obs}}$. The local ray residual is defined as

$$\mathbf{r}_{t,b,j}^{\text{ray}} = \mathbf{B}_{t,b,j}^\top \left(\mathbf{d}_{t,b,j}^{\text{pred}} - \mathbf{d}_{t,b,j}^{\text{obs}} \right). \quad (13)$$

This two-dimensional residual preserves the tangential degrees of freedom of the ray discrepancy and provides a first-order approximation of angular error in the small-residual regime. The observed ray and its tangent basis are updated together with the camera parameters during optimization.

We additionally define a normalized hybrid residual that balances the pixel reprojection residual $\mathbf{r}_{t,b,j}^{\text{px}}$ and the ray residual:

$$\mathbf{r}_{t,b,j}^{\text{hyb}} = \begin{bmatrix} \sqrt{1-\lambda} \mathbf{r}_{t,b,j}^{\text{px}} / s_{\text{px}} \\ \sqrt{\lambda} \mathbf{r}_{t,b,j}^{\text{ray}} / s_{\text{ray}} \end{bmatrix}, \quad \lambda \in [0, 1], \quad (14)$$

where s_{px} and s_{ray} are fixed normalization scales computed from the median residual magnitudes before final refinement. The standard pixel objective is used by default, while the ray-domain and hybrid objectives are provided as optional final refinement alternatives.

IV. EXPERIMENT

A. Experimental Setup

B. Ablation

This section evaluates two aspects of the recovered internal lattice observations: their generation reliability and their constraining effect on camera calibration. We first compare internal-seed constructions in different geometric domains, and then examine whether the recovered internal observations improve intrinsic recovery.

1) *Internal Seed Generation*: Given accurate outer-corner detections, internal-point refinement depends on the geometric quality of the initial seeds. We compare two strategies: Virtual Pinhole Patch and Spherical Seed. Following TartanCalib, Virtual Pinhole Patch uses an intermediate camera model to remap the wide-FoV image

TABLE I

CROSS-BACKEND VERIFICATION ON THE HELD-OUT MULTI-BOARD AND INDEPENDENT CHECKERBOARD TEST SETS. BABELCALIB USES THE SAME DETECTED OBSERVATIONS AND TARGET GEOMETRY AS OURS.

Model	Method / Observations	f_u/f_v	c_u/c_v	Model Params.	Test RMSE [px] ↓	
					Multi-board	Checkerboard
DS	Ours	1175.65 / 1175.29	2242.88 / 2275.61	$[-0.1905, 0.6171]$	0.5228	0.5918
	BabelCalib w/ Outer+Internal	1176.56 / 1175.92	2242.90 / 2275.63	$[-0.1914, 0.6175]$	0.5293	0.6020
	BabelCalib w/ Outer4 only	1943.49 / 1943.08	2242.91 / 2275.67	$[0.3401, 0.8048]$	0.5256 [†]	–
KB	Ours	1451.58 / 1451.10	2242.37 / 2275.38	$[-0.0197, 0.0010, -0.0032, 0.0006]$	0.5216	0.5960
	BabelCalib w/ Outer+Internal	1454.52 / 1453.74	2242.91 / 2275.65	$[-0.0196, -0.0018, -0.0007, -0.0001]$	0.5286	0.6052
	BabelCalib w/ Outer4 only	–	–	–	–	–

[†] The Outer4-only setting achieves a comparable multi-board test RMSE but converges to a substantially shifted intrinsic parameterization.

TABLE II

INTERNAL-SEED ABLATION WITH A SHARED OUTER4 INITIALIZATION.

Method	Reproj. Err. [px] ↓	Outer Err. [px] ↓	Internal Err. [px] ↓	Max. Err. [px] ↓
Outer-only	0.5251	0.1002	0.5582	1.679
Virtual Pinhole Patch	4.3647	0.1089	4.6499	50.353
Spherical Seed	0.5243	0.1030	0.5572	1.678

Note. Spherical Seed is our method. Internal error uses frozen validation points excluded from Outer-only optimization; max. error is the maximum frame–board RMSE.

into virtual pinhole views. In contrast, Spherical Seed constructs the seeds directly in the viewing-ray domain via spherical interpolation. Both variants use the same Outer4 initialization, local refinement, and optimization backend. We also include Outer-only as a baseline without internal-point recovery.

Table II summarizes the results. Virtual Pinhole Patch achieves an outer error comparable to Spherical Seed, but produces substantially larger internal and maximum errors, increasing the overall reprojection error. Since the two variants share the same Outer4 initialization, the degradation mainly arises from unreliable internal seeds. In contrast, Spherical Seed maintains an overall error comparable to Outer-only while incorporating additional internal observations, indicating that viewing-ray-domain seed construction is more consistent with wide-FoV projection geometry.

2) *Internal Observation Evaluation:* The preceding experiment shows that Spherical Seed can reliably recover internal observations. However, because it shares the same Outer4 initialization with Outer-only, both settings start from a nearly converged model. Their final reprojection errors therefore cannot isolate the contribution of internal lattice points to intrinsic recovery. To directly evaluate this contribution, we construct a controlled intrinsic-recovery experiment using the Double Sphere (DS) model. Both settings share the same reference scene, perturbed initialization, Outer4 frame–board observations, and paired noise realizations; Outer+Internal additionally incorporates only the internal lattice points

recovered by Spherical Seed.

In wide-FoV calibration, limited radial coverage may cause focal scale and projection-shape parameters to induce similar residual changes. Rather than arbitrarily perturbing an individual intrinsic parameter, we extract the least-constrained intrinsic direction $\mathbf{d}_{W_1}$ from the pose-eliminated Schur information matrix computed using the full-budget Outer-only observations. This direction is fixed for both settings and across all observation budgets. For method m and budget b , the information along this direction is defined as

$$I_{W_1}^{m,b} = \mathbf{d}_{W_1}^\top \mathbf{S}_c^{m,b} \mathbf{d}_{W_1}. \quad (15)$$

We then perturb the reference DS intrinsics along this direction such that the initial model has a P95 ray error of 2° in the peripheral region $\rho \geq 0.7$. Fig. 2 evaluates the information along the weakest intrinsic direction, the model-recovery error over the full FoV, and the observation budget required to reach the prescribed peripheral-accuracy target.

We then perturb the reference DS intrinsics along $\mathbf{d}_{W_1}$ such that the initial model reaches a P95 ray error of 2° in the peripheral region $\rho \geq 0.7$. Fig. 2 compares the information along the weakest intrinsic direction, the model-recovery error over the full FoV, and the observation budget required to reach the prescribed peripheral-accuracy target.

As shown in Fig. 2(a), incorporating internal lattice points consistently strengthens the constraint along the weakest intrinsic direction across all frame–board budgets. Fig. 2(b) shows that the recovered model more closely matches the reference model over the full FoV, with larger improvements near the image periphery. Fig. 2(c) further shows that Outer+Internal reaches the prescribed peripheral-accuracy target with fewer observations. These results indicate that internal lattice points do not replace the initialization provided by spatially distributed Outer4 observations. Instead, they provide complementary local constraints that improve DS model recovery in highly distorted peripheral regions.

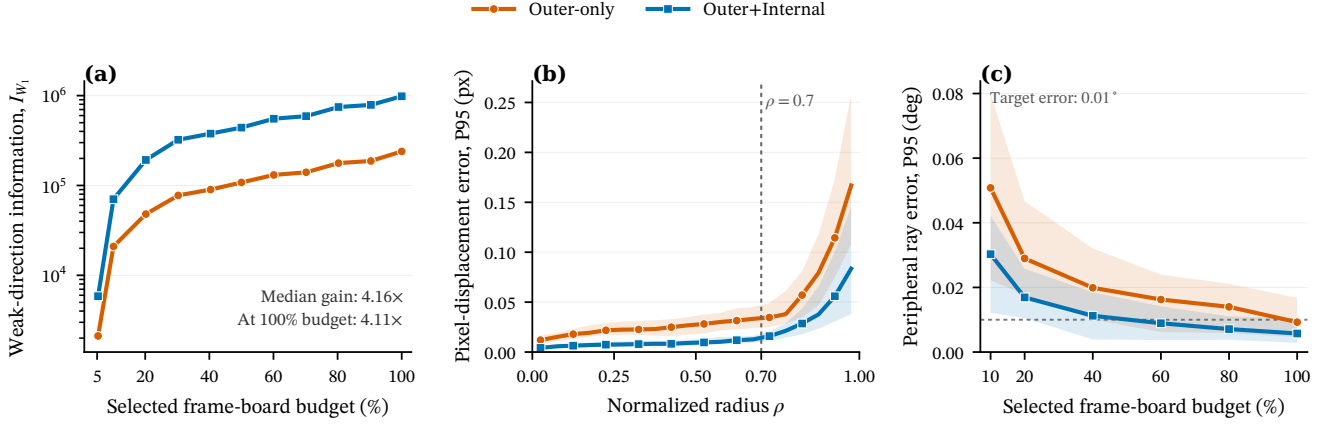


Fig. 2. Effect of internal observations on the recovery of perturbed DS intrinsics. Outer-only and Outer+Internal share the same Outer4 frame-board observations, perturbed initialization, and paired noise realizations; Outer+Internal additionally incorporates the recovered internal lattice points. (a) Schur information along the least-constrained intrinsic direction W_1 after pose elimination. (b) P95 radial pixel displacement between the recovered and reference models, evaluated on a fixed grid of reference rays. (c) Peripheral P95 ray error under different frame-board budgets. Both settings start from the same 2° peripheral perturbation and optimize only the DS intrinsics. Panels (b) and (c) summarize 100 paired trials; the curves and shaded regions denote the median and interquartile range (IQR), respectively. The peripheral region is defined as $\rho \geq 0.7$.

C. Camera Pose Estimation

To evaluate the calibration results, we perform camera pose estimation on held-out test images. The images are randomly split into training and test sets at a ratio of 7:3. For each method, we fix the evaluated camera intrinsics and the shared multi-board geometry during testing, and optimize only the pose of the reference board with respect to the camera by minimizing a robust reprojection objective. Reprojection errors are then evaluated over all visible board observations in each image using RMSE, P95, and Inlier@1px.

Since the compared calibration pipelines are not directly applicable to our multi-board calibration setup, we estimate their camera intrinsics using standard checkerboard-based calibration procedures. The resulting intrinsics are then evaluated alongside ours on the same held-out multi-board pose-estimation task and are reported as reference baselines.

V. CONCLUSIONS

A conclusion section is not required. Although a conclusion may review the main points of the paper, do not replicate the abstract as the conclusion. A conclusion might elaborate on the importance of the work or suggest applications and extensions.

APPENDIX

Appendixes should appear before the acknowledgment.

ACKNOWLEDGMENT

The preferred spelling of the word *Acknowledgment* in America is without an *e* after the *g*. Avoid the stilted expression, *One of us (R. B. G.) thanks . . .* Instead, try *R. B. G. thanks*. Put sponsor acknowledgments in the unnumbered footnote on the first page.

References are important to the reader; therefore, each citation must be complete and correct. If at all possible, references should be commonly available publications.

REFERENCES

- [1] V. Usenko, N. Demmel, and D. Cremers, "The double sphere camera model," in *2018 International Conference on 3D Vision (3DV)*. IEEE, 2018, pp. 552–560.